complex variable, *s*, with its equivalent expression, $\sigma + j\omega$. Using this alternate notation, the Laplace transform becomes:

$$X(\sigma,\omega) = \int_{t=-\infty}^{\infty} x(t)\, e^{-\sigma t} e^{-j\omega t}\, dt$$

If we are only concerned with *real* time domain signals (the usual case), the top and bottom halves of the s-plane are mirror images of each other, and the term, $e^{-j\omega t}$, reduces to simple cosine and sine waves. This equation identifies each *location* in the s-plane by the two parameters, σ and ω. The *value* at each location is a complex number, consisting of a real part and an imaginary part. To find the real part, the time domain signal is multiplied by a cosine wave with a frequency of ω, and an amplitude that changes exponentially according to the decay parameter, σ. The value of the real part of $X(\sigma,\omega)$ is then equal to the integral of the resulting waveform. The value of the imaginary part of $X(\sigma,\omega)$ is found in a similar way, except using a sine wave. If this doesn't sound very familiar, you need to review the previous chapter before continuing.

The Laplace transform can be changed into the z-transform in three steps. The first step is the most obvious: change from continuous to discrete signals. This is done by replacing the time variable, *t*, with the sample number, *n*, and changing the integral into a summation:

$$X(\sigma,\omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-\sigma n} e^{-j\omega n}$$

Notice that $X(\sigma,\omega)$ uses parentheses, indicating it is *continuous*, not discrete. Even though we are now dealing with a discrete time domain signal, $x[n]$, the parameters σ and ω can still take on a continuous range of values. The second step is to rewrite the exponential term. An exponential signal can be mathematically represented in either of two ways:

$$y[n] = e^{-\sigma n} \quad \text{or} \quad y[n] = r^n$$

As illustrated in Fig. 33-1, both these equations generate an exponential curve. The first expression controls the decay of the signal through the parameter, σ. If σ is positive, the waveform will *decrease* in value as the sample number, *n*, becomes larger. Likewise, the curve will progressively *increase* if σ is negative. If σ is exactly zero, the signal will have a constant value of *one*.